

Course 5: Mathematical Models for Vehicle Routing Problems

Doc-Course Optimization over Nonlinear Model Spaces:
Where Discrete Meets Continuous Optimization (ONMoS)

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- 1 Problem description
- 2 Compact formulations
- 3 Cutting-plane formulations
- 4 Set-partitioning formulations
- 5 Comparing formulations

Chapter 7 “The Capacitated Vehicle Routing Problem” (D.Vigo & J.J.Salazar)
from book “Handbook on Transport Modeling” (S.Parragh & T.VanWoensel)
Series “Handbooks in Transport Studies”, Edward Elgar Publishing Ltd, 2026

Outline

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Motivation

- “Vehicle Routing Problems” was the topic of the **12th DIMACS Implementation Challenge**.
- Benchmark collection of instances with 100 to 1000 customers:
E.Uchoa, D.Pecin, A.Pessoa, M.Poggi, T.Vidal, A.Subramanian, New benchmark instances for the Capacitated Vehicle Routing Problem, European Journal of Operational Research 257 (2017) 845–858.
- The best exact solution approach is a branch-and-cut-and-price algorithm:
D. Pecin, A. Pessoa, M. Poggi, E. Uchoa, Improved branch-cut-and-price for capacitated vehicle routing, LNCS 8494 (2014) 393-403
This approach solves most instances with up to **275** customers, but not all. For example, it did not solve 38 benchmark instances in that range. It did solve 8 larger instances (with short routes or small vehicle capacities).
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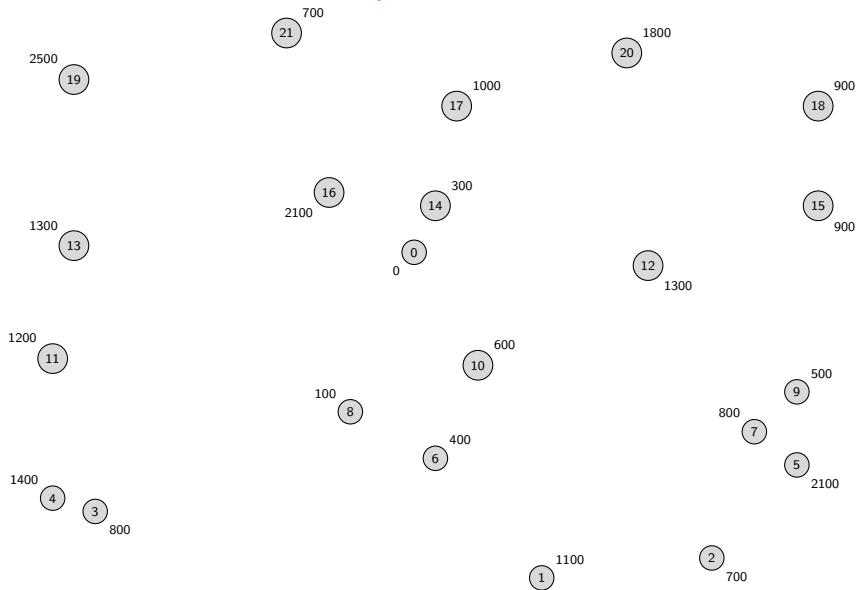
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Background

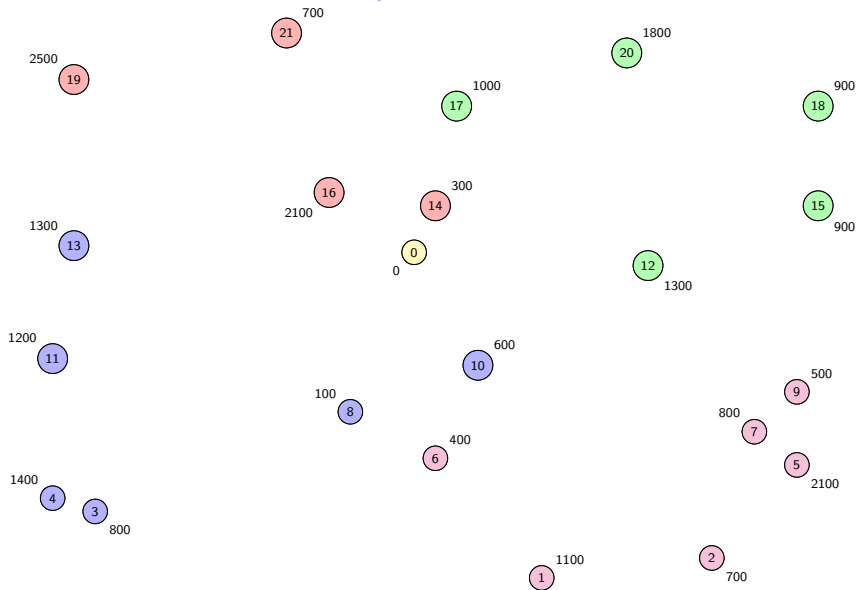
- Customers requiring service (or tasks to performs): can be mandatory or optional, perhaps with estimated demands
- Vehicles to execute the service (or machines), perhaps with capacities: can all identical or heterogeneous fleet
- Known travel costs (distance or time). Stochastic data in some cases.
- There may be time-window constraints, precedence constraints, one or several visits to each customer,...
- There may be different products to deliver to or collect from customers.
- There may be one or several depots
- Some variants require using “all” available vehicles, and other “at most” the available vehicles.
- The aim is to design the vehicle routes, which means two subproblems: **CLUSTER** (partitioning) and **ROUTE** (permuting)
- The objective could be the minimization of the sum (or maximum, or the difference between the maximum and the minimum) route costs
- Google Scholar reports about 270,000 documents on “vehicle routing”, meaning the importance of the topic and the wide problem variety.

Instance E-n22-k4: 1 depot, $n = 21$, $m = 4$, $Q = 6000$.



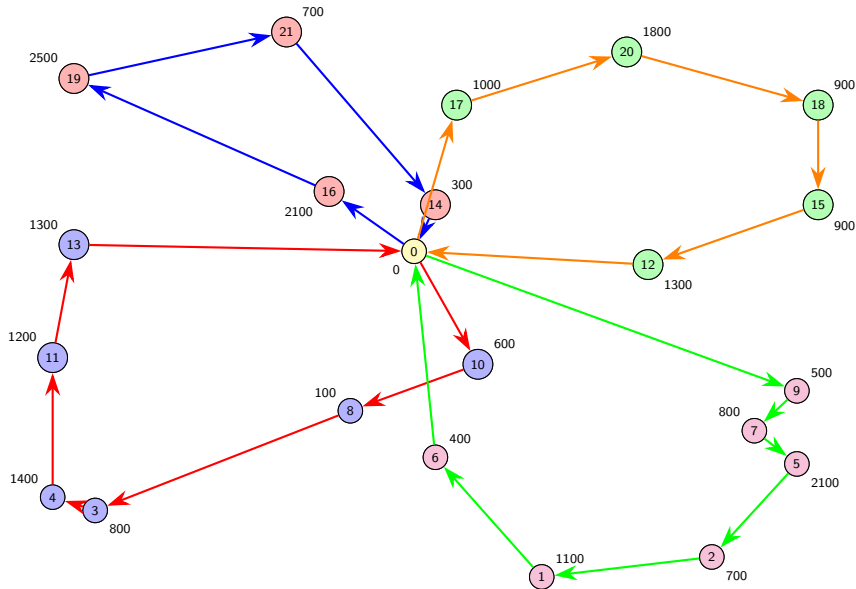
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How should we **CLUSTER** and **ROUTE** to minimize the total routing cost?

Notation

- $G = (V, A)$ directed graph with $V = \{0, 1, \dots, n\}$ the vertex set.
- 0 represents the **depot**. $N = \{1, \dots, n\}$ represent the **customers**.
- m vehicles, all with identical vehicle **capacity** Q .
- $K = \{1, \dots, m\}$ indices to determine the homogeneous fleet of vehicles.
- customer **demands** [delivery to customer]: $0 \leq q_i \leq Q$ (and say $q_0 = 0$).
- $\forall S \subseteq N$, we write $q(S)$ instead of $\sum_{i \in S} q_i$.
- $\forall S \subseteq N$, let $r(S)$ = be minimum number of bins of size Q to pack items in S .
- Let $m^* = r(N)$. Thus, $q(N)/Q \leq m^*$. We assume $m^* \leq m$.
- Let $c = [c_a : a \in A]$ be a given matrix. We use $c_a = c_{ij}$ when $a = (i, j)$.
- We do not assume properties on this matrix.
i.e. our results work also when e.g. $c_{ij} \neq c_{ji}$ or $c_{ij} + c_{jk} < c_{i,k}$.
- Standard notation in Graph Theory: $\forall S \subseteq V$,
 - ▶ $\delta^+(S) = \{(i, j) \in A : i \in S, j \notin S\}$
 - ▶ $\delta^-(S) = \{(i, j) \in A : i \notin S, j \in S\}$
 - ▶ $A(S) = \{(i, j) \in A : i \in S, j \in S\}$
- $\forall i \in V$, we write i instead of $\{i\}$.

CLUSTER and ROUTE simultaneously

CVRP looks for a minimum-cost set of routes to server all customers' demands subject to

- each customer is visited once (service is not optional)
- each vehicle starts and end at the depot (routes are simple circuits)
- for each vehicle, the sum of served demands is not larger than Q

Thus, the **CVRP** combines two classical combinatorial problems:

- Bin Packing Problem (BPP)
- Travelling Salesman Problem (TSP)

CVRP is TSP when $m = 1$ and $Q = \infty$, thus **strongly NP-hard**.

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$$x_{ij}^k = \begin{cases} 1 & \text{if } (i,j) \text{ in the route for vehicle } k \\ 0 & \text{otherwise} \end{cases} \quad \text{for all } (i,j) \in A \text{ and } k \in K.$$
$$y_i^k = \begin{cases} 1 & \text{if } i \text{ served by vehicle } k \\ 0 & \text{otherwise} \end{cases} \quad \text{for all } i \in N \text{ and } k \in K.$$

$$\min \sum_{a \in A} c_a \sum_{k \in K} x_a^k$$

subject to

$$\sum_{k \in K} y_i^k = 1 \quad \forall i \in N$$

$$\sum_{a \in \delta^+(i)} x_a^k = \sum_{a \in \delta^-(i)} x_a^k = y_i^k \quad \forall k \in K, i \in N$$

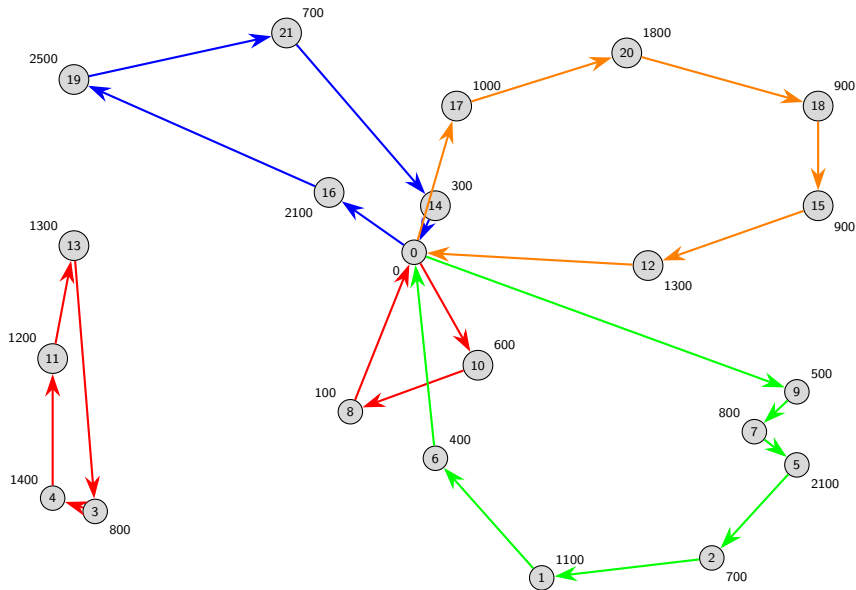
$$\sum_{a \in \delta^+(0)} x_a^k = \sum_{a \in \delta^-(0)} x_a^k = 1 \quad \forall k \in K$$

$$\sum_{i \in N} q_i y_i^k \leq Q \quad \forall k \in K$$

[no subtours in N]

$$y_i^k \in \{0, 1\} \quad \forall i \in N, k \in K$$

$$x_a^k \in \{0, 1\} \quad \forall a \in A, k \in K.$$



We do not want SUBTOURS within customers!!!!

3-index (x_{ij}^k)

There are many ways to force [no subtours in N] through linear inequalities inspired from TSP formulations.

A **compact** example is based in the Miller-Tucker-Zemlin's TSP formulation:

$$u_j \geq u_i + x_{ij}^k - (|N| - 1)(1 - x_{ij}^k) + (|N| - 2)x_{ji}^k \quad \forall i, j \in N : (i, j) \in A, k \in K$$

where u_i is a continuous variable for each $i \in N$ representing the relative position of i in a vehicle route.

A **non-compact** example is based on extending the subtour elimination constraints:

$$\sum_{a \in \delta^+(S)} x_a^k \geq y_i^k \quad \forall i \in N, S \subset N : i \in S, k \in K$$

Advantage: no additional variables, stronger LP relaxation

Disadvantage: It needs a cutting-plane framework.

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$$x_a = \begin{cases} 1 & \text{if } a \text{ in the route of a vehicle} \\ 0 & \text{otherwise} \end{cases} \quad \forall a \in A$$

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[vehicle capacity limits]

[no subtours in N]

$$x_a \in \{0, 1\} \quad \forall a \in A.$$

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2-index (x_{ij}): Miller-Tucker-Zemlin

u_i representing the amount of product delivered after visiting customer i , $\forall i \in N$
[warning: the load of a vehicle is decreasing along the route, and u_i is *increasing*]

$$\begin{aligned} u_j &\geq u_i + q_j - Q(1 - x_{ij}) & \forall (i,j) \in A \mid i,j \in N \\ q_i &\leq u_i \leq Q & \forall i \in N \end{aligned}$$

or better with

$$\begin{aligned} u_j &\geq u_i + q_j - Q(1 - x_{ij}) + (Q - q_i - q_j)x_{ji} & \forall (i,j) \in A \mid i,j \in N \\ q_i + \sum_{j \in N \setminus \{i\}} q_j x_{ji} &\leq u_i \leq Q - \sum_{j \in N \setminus \{i\}} q_j x_{ij} & \forall i \in N. \end{aligned}$$

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2-index (x_{ij}): adapted to heterogeneous fleet

Still using u_i variables, also consider:

$$z_i^k = \begin{cases} 1 & \text{if } i \text{ is the last customer served by vehicle } k \\ 0 & \text{otherwise} \end{cases} \quad \forall i \in N \text{ and } k \in K.$$

[warning: it is **last** because u_i is *increasing*, hence we must limit u_i when $x_{i0} = 1$]

$$\sum_{i \in N} z_i^k = 1 \quad \forall k \in K$$

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$$q_i \leq u_i \leq \sum_{k \in K} Q^k z_i^k + Q(1 - x_{i0}) \quad \forall i \in N$$

where $Q = \max\{Q^k \mid k \in K\}$.

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$$\begin{aligned} \sum_{i \in N} z_i^k &= 1 & \forall k \in K \\ \sum_{k \in K} z_i^k &= x_{i0} & \forall i \in N \\ z_i^k &\in \{0, 1\} & \forall i \in N, k \in K \\ q_i \leq u_i &\leq \sum_{k \in K} Q^k z_i^k + Q(1 - x_{i0}) & \forall i \in N \end{aligned}$$

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2-index (x_{ij}) with single commodity flow variables

By Gavish&Graves(1979):

f_a representing the load of a vehicle traversing arc a , $\forall a \in A$
[warning: the load f_a is *decreasing* along each route!]

$$\begin{aligned}\sum_{a \in \delta^-(i)} f_a - \sum_{a \in \delta^+(i)} f_a &= q_i & \forall i \in N \\ f_a &\geq 0 & \forall a \in A \\ f_{0i} &\leq Qx_{0i} & \forall i \in N\end{aligned}$$

or better with

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$$\sum_{k \in K} z_i^k = x_{0i} \quad \forall i \in N$$

$$z_i^k \in \{0, 1\} \quad \forall i \in N, k \in K$$

$$f_{0i} \leq \sum_{k \in K} Q^k z_i^k \quad \forall i \in N.$$

2-index (x_{ij}) with two commodities flow variables

By Baldacci et al. (1999)

While f_a determines a flow delivering a commodity along the route determined by x_a , the value $Q - f_a$ can be seen as another flow collecting a different product (e.g., the empty space in the vehicle). Each flow delivers or collects q_i units when passing through customer i . Inequality $f_{ij} \leq Qx_{ij}$ forces $f_{ji} = 0$ when $x_{ij} = 1$, thus only the delivery flow is reflected with the f variables.

$$\begin{aligned} \sum_{a \in \delta^-(i)} h_a - \sum_{a \in \delta^+(i)} h_a &= 2q_i & \forall i \in N \\ h_{ij} + h_{ji} &= Q(x_{ij} + x_{ji}) & \forall i, j \in N \mid i < j \\ h_a &\geq 0 & \forall a \in A. \end{aligned}$$

This model is equivalent to the previous single-commodity flow formulation.

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2-index (x_{ij}) with a partial precedence relation

$$v_{ij} = \begin{cases} 1 & \text{if } i \text{ served before } j \text{ by the same vehicle} \\ 0 & \text{otherwise} \end{cases} \quad \forall (i,j) \in A$$

$$\begin{aligned} v_{ij} + v_{ji} &\leq 1 & \forall i,j \in N | i \neq j \\ v_{ij} + v_{jk} &\leq 1 + v_{ik} & \forall i,j,k \in N | i \neq j, i \neq k, j \neq k \\ x_{ij} &\leq v_{ij} & \forall (i,j) \in A \end{aligned}$$

and

$$\sum_{j \in N \setminus \{i\}} q_j (v_{ji} + v_{ij}) \leq Q - q_i \quad \forall i \in N$$

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2-index (x_{ij}) with one multi commodity flow variables

By Garvin et al (1957) ; it includes 3-index variables:

$$f_{ij}^k = \begin{cases} 1 & \text{if a vehicle travels } (i,j) \text{ from depot to } k \\ 0 & \text{otherwise} \end{cases} \quad \forall (i,j) \in A \text{ and } k \in N.$$

$$\sum_{a \in \delta^+(0)} f_a^k = \sum_{a \in \delta^-(k)} f_a^k = 1 \quad \forall k \in N$$

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2-index (x_{ij}) with one multi commodity flow variables

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$$\sum_{k \in N} q_k f_a^k \leq Q x_a \quad \forall a \in A. \quad (1)$$

The LP relaxation gives a better bound when adding

$$f_a^k \leq x_a \quad \forall k \in N, a \in A. \quad (2)$$

An alternative to inequalities (1) is

$$\sum_{k \in N \setminus \{i\}} q_k \sum_{a \in \delta^+(i)} f_a^k \leq Q - q_i \quad \forall i \in N \quad (3)$$

and now (2) is mandatory to have a valid CVRP model.

No dominance relation exist.

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2-index (x_{ij}) with one multi commodity flow variables

Inequalities (1)

$$\sum_{k \in N} q_k f_a^k \leq Q x_a \quad \forall a \in A.$$

can be rewritten as

$$\sum_{k \in N \setminus \{j\}} q_k f_{ij}^k + q_j f_{ij}^j \leq Q x_{ij} \quad \forall (i, j) \in A.$$

Since $f_{ij}^j = x_{ij}$, the inequality can be rewritten as

$$\sum_{k \in N \setminus \{j\}} q_k f_{ij}^k \leq (Q - q_j) x_{ij} \quad \forall (i, j) \in A.$$

Since $f_{ij}^i = 0$, a strengthening is obtained by also reducing Q with q_i as follows:

$$\sum_{k \in N \setminus \{i, j\}} q_k f_{ij}^k \leq (Q - q_i - q_j) x_{ij} \quad \forall (i, j) \in A.$$

2-index (x_{ij}) with one multi commodity flow variables

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Inequalities (3)

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can be strengthened by

$$\sum_{k \in N \setminus \{i\}} q_k \left(\sum_{a \in \delta^+(i)} f_a^k + \sum_{a \in \delta^+(k)} f_a^i \right) \leq Q - q_i \quad \forall i \in N.$$

because $\sum_{a \in \delta^+(i)} f_a^k = 1$ means that both i and k are served by the same vehicle.
Also $\sum_{a \in \delta^+(k)} f_a^i = 1$ means that both i and k are served by the same vehicle.

Even more, $\sum_{a \in \delta^+(i)} f_a^k$ defines a **partial precedence relation** between customers.
Thus $\sum_{a \in \delta^+(i)} f_a^k$ leads to valid inequalities like

$$\sum_{a \in \delta^+(i)} f_a^k + \sum_{a \in \delta^+(k)} f_a^i \leq 1 \quad \forall i, k \in N \mid i \neq k$$

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$$\sum_{a \in \delta^+(i)} f_a^j + \sum_{a \in \delta^+(j)} f_a^k \leq 1 + \sum_{a \in \delta^+(i)} f_a^k \quad \forall i, j, k \in N \mid i \neq j, i \neq k, j \neq k.$$

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2-index (x_{ij}) two multi commodity flow variables

By Letchford & Salazar (2015):

$$g_a^k = \begin{cases} 1 & \text{a vehicle traverses } a \text{ on the way from } k \text{ to depot} \\ 0 & \text{otherwise} \end{cases} \quad \forall a \in A \text{ and } k \in V$$

$$\sum_{a \in \delta^+(k)} g_a^k = \sum_{a \in \delta^-(0)} g_a^k = 1 \quad \forall k \in N$$

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$$\sum_{k \in N \setminus \{i\}} q_k \left(\sum_{a \in \delta^+(k)} f_a^i + \sum_{a \in \delta^-(k)} g_a^i \right) \leq Q - q_i \quad \forall i \in N$$
$$f_a^k + g_a^k \leq x_a \quad \forall k \in N, a \in A.$$

It can be strengthened by also considering

$$\sum_{a \in \delta^+(i)} f_a^k = \sum_{a \in \delta^+(k)} g_a^i \quad \forall k, i \in N \mid i \neq k.$$

A further strengthening is with

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2-index (x_{ij}) two multi commodity flow variables

To reduce the number of variables, set $h_{ij}^k = f_{ij}^k + g_{ji}^k$:

$$\sum_{a \in \delta^+(0)} h_a^k = \sum_{a \in \delta^-(k)} h_a^k = 2 \quad \forall k \in N$$

$$\sum_{a \in \delta^-(0)} h_a^k = \sum_{a \in \delta^+(k)} h_a^k = 0 \quad \forall k \in N$$

$$\sum_{a \in \delta^+(i)} h_a^k = \sum_{a \in \delta^-(i)} h_a^k \quad \forall k, i \in N \mid i \neq k$$

$$h_a^k \geq 0 \quad \forall k \in N, a \in A$$

and

$$\sum_{k \in N \setminus \{i,j\}} q_k (h_{ij}^k + h_{ji}^k) \leq (Q - q_i - q_j)(x_{ij} + x_{ji}) \quad \forall i, j \in N \mid i < j$$

$$h_{ij}^k + h_{ji}^k \leq x_{ij} + x_{ji} \quad \forall k \in N, i, j \in V \mid i < j$$

$$\sum_{a \in \delta^+(i)} h_a^k = \sum_{a \in \delta^+(k)} h_a^i \quad \forall k, i \in N \mid i < k.$$

Outline

- 1 Problem description
- 2 Compact formulations
- 3 Cutting-plane formulations**
- 4 Set-partitioning formulations
- 5 Comparing formulations

Inequalities on the 2-index variables x_{ij}

- The *subtour elimination* (SE):

$$\sum_{a \in \delta^+(S)} x_a \geq 1 \quad \forall S \subseteq N$$

- The *bin packing* (BP):

$$\sum_{a \in \delta^+(S)} x_a \geq r(S) \quad \forall S \subseteq N$$

- The *fractional capacity* (FC):

$$\sum_{a \in \delta^+(S)} x_a \geq \frac{q(S)}{Q} \quad \forall S \subseteq N$$

- The *rounded capacity* (RC):

$$\sum_{a \in \delta^+(S)} x_a \geq \left\lceil \frac{q(S)}{Q} \right\rceil \quad \forall S \subseteq N$$

Inequalities on the 2-index variables x_{ij}

- The *generalized large multistar* (GLM):

$$\sum_{a \in \delta^+(S)} x_a \geq \frac{1}{Q} \sum_{i \in S} (q_i + \sum_{j \in N \setminus S} q_j (x_{ij} + x_{ji})) \quad \forall S \subseteq N$$

- The *knapsack large multistar* (KLM):

$$\sum_{a \in \delta^+(S)} x_a \geq \frac{1}{\beta} \sum_{i \in S} (\alpha_i + \sum_{j \in N \setminus S} \alpha_j (x_{ij} + x_{ji})) \quad \forall S \subseteq N$$

where $\alpha \geq 0$ and $\beta > 0$ are such that $\sum_{i \in N} \alpha_i y_i \leq \beta$ valid for

$$KP(Q, q) := \text{conv} \{y \in \{0, 1\}^n : \sum_{i \in N} q_i y_i \leq Q\}.$$

Managing a large number of inequalities

A family $\mathcal{F}$ with an exponential number of valid inequalities requires solving the

Separation Problem

Find an inequality in $\mathcal{F}$ maximizing the violation of given primal variables x^* .

With a **separation algorithm**, the model can be solved in an iterative *row-generation* framework which at each iteration solves

- a *master problem* (with some inequalities in $\mathcal{F}$) using a general-purpose solver
- a *subproblem* (the separation problem) with the separation algorithm

We call it *cutting-plane phase*, or simply *cutting*.

In general, solving these models imply dealing with two difficulties:

- *cutting* to manage the large number of inequalities
- *branching* to force some variables to be integer

Both are *iterative procedures* (i.e. implemented within a `while...do...`)

Two extreme (and common) implementation options:

- **cut-and-branch**: separation problem only on integer solutions x^*
- **branch-and-cut**: separation problem on any (also fractional) solution x^*

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Separation Procedures

The separation approaches for some basic CVRP inequalities $\mathcal{F}$ make use of the

Max-flow Min-cut Theorem (Hoffman 1974)

Let us consider a graph $G' = (V', A')$, two nodes $s, t \in V'$, and each $a \in A'$ with lower and upper capacities $l_a \leq u_a$. Finding $S \subset V' : s \in S, t \notin S$ minimizing

$$\sum_{a \in \delta^+(S)} u_a - \sum_{a \in \delta^-(S)} l_a$$

is equivalent to solving the *Maximum Flow Problem* from s to t (polynomial-time)

The method consists of rewriting $\mathcal{F}$ to have a constant in the right-hand side, and then build G' and l, u such that a cut capacity is the left-hand side.

Separating SE inequalities

$$\sum_{a \in \delta^+(S)} x_a \geq 1 \quad \forall S \subseteq N$$

It is direct application of the Max-flow Min-cut Theorem:

For each $i \in N$, we must solve one Maximum Flow Problem.

$G' = G, V' = V, A' = A, s = 0, t = i, l_a = 0, u_a = x_a^*$ the given solution.

There is a violated SE inequality iff the maximum 0- i flow capacity < 1 .

Equivalent to compute the max-flow from 0 to i with

$$0 \leq f_a \leq 1 \quad \forall a \in A$$

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$$0 \leq f_a \leq 1 \quad \forall a \in A$$

Separating FC inequalities

$$\sum_{a \in \delta^+(S)} x_a \geq \frac{q(S)}{Q} \quad \forall S \subseteq N$$

It needs a slight modification of the previous approach:

The FC inequalities can be rewritten as

$$\sum_{a \in \delta^+(S)} Qx_a + \sum_{j \notin S} q_j \geq q(N) \quad \forall S \subseteq N$$

$$V' = V \cup \{n+1\}, s = 0, t = n+1, A' = A \cup \{(i, n+1) : i \in N\} \\ l_a = 0, u_a = Qx_a^* \text{ for all } a \in A; l_{i,n+1} = 0, u_{i,n+1} = q_i \text{ for all } i \in N$$

There is a violated FC inequality iff the maximum 0-($n+1$) flow capacity $< q(N)$.

Equivalent to compute the max-flow from 0 to $n+1$ with

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is **NP-hard**; see I. Diarrassouba, "On the complexity of the separation problem for rounded capacity inequalities", Discrete Optimization 25 (2017) 86–104.

Still, we may implement a separation procedure! Let see an exact one:

Given x^* , for each $k \in \{1, \dots, m-1\}$, the separation problem is equivalent to checking whether there S requiring more than k vehicles but $\sum_{a \in \delta^+(S)} x_a^* < k + 1$.

It is equivalent to finding a **min cut** $(S, V \setminus S)$ **subject to** $q(S) > kQ$.

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$$y_i = \begin{cases} 1 & \text{if } i \in S \\ 0 & \text{otherwise} \end{cases} \quad \forall i \in N$$

$$w_{ij} = \begin{cases} 1 & \text{if } i \in S, j \notin S \\ 0 & \text{otherwise} \end{cases} \quad \forall (i,j) \in A$$

For each $k \in \{1, \dots, m-1\}$:

$$\begin{aligned} & \min \sum_{(i,j) \in A} x_{ij}^* w_{ij} \\ & \text{subject to:} \quad y_0 = 0 \\ & \quad w_{ij} \geq y_i - y_j \quad \forall (i,j) \in A \\ & \quad w_{ij} \geq 0 \quad \forall (i,j) \in A \\ & \quad \sum_{i \in N} q_i y_i \geq k \cdot Q + 1 \\ & \quad y_i \in \{0, 1\} \quad \forall i \in N \end{aligned}$$

K. Pavlikov, N.C. Petersen, J.L. Sørensen, "Exact separation of the rounded capacity inequalities for the capacitated vehicle routing problem", *Networks* 83(2024)197–209

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$$\sum_{a \in \delta^+(S)} x_a \geq r(S) \quad \forall S \subseteq N$$

is **NP-hard** because only computing $r(S)$ for a given S is.

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$$r_k = \begin{cases} 1 & \text{if bin } k \text{ includes a customer in } S \\ 0 & \text{otherwise} \end{cases} \quad \forall i \in N$$

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$$0 \leq z_i^k \leq r_k \quad \forall i \in N, k \in K$$

$$\sum_{i \in N} q_i z_i^k \leq Q r_k \quad \forall k \in K$$

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Bilevel Optimization: There is a violated BP inequality iff the opt obj val < 0 .

Separating **stronger** BP inequalities

To also pack customers outside S with (at most) m vehicles:

$$\begin{aligned} \min \quad & \sum_{(i,j) \in A} x_{ij}^* w_{ij} - \sum_{k \in K} r_k \\ \text{subject to:} \quad & y_0 = 0 \\ & w_{ij} \geq y_i - y_j & \forall (i,j) \in A \\ & w_{ij} \geq 0 & \forall (i,j) \in A \\ & y_i \in \{0,1\} & \forall i \in N \\ \min \quad & \sum_{k \in K} r_k \\ \text{subject to} \quad & \sum_{k \in K} z_i^k = 1 & \forall i \in N \\ & \sum_{i \in N} q_i z_i^k \leq Q & \forall k \in K \\ & y_i + z_i^k \leq 1 + r_k & \forall i \in N, k \in K \\ & z_i^k \geq 0 & \forall i \in N, k \in K \\ & r_k \in \{0,1\} & \forall k \in K. \end{aligned}$$

Outline

- 1 Problem description
- 2 Compact formulations
- 3 Cutting-plane formulations
- 4 Set-partitioning formulations**
- 5 Comparing formulations

Set partitioning (SP): Balinski-Quandt (1964)

Let R be the collection of all *CVRP feasible* routes, and for each $r \in R$:

Let a_i^r be the number of times i visited in r , $\forall i \in N$

Let b_a^r be the number of times a traversed in r , $\forall a \in A$

Let $c_r = \sum_{a \in A} b_a^r c_a$

$$z_r = \begin{cases} 1 & \text{if } r \text{ selected} \\ 0 & \text{otherwise} \end{cases} \quad \forall r \in R$$

$$\min \sum_{r \in R} c_r z_r$$

$$\text{s.t.} \quad \sum_{r \in R} a_{ir} z_r = 1 \quad \forall i \in N \quad (4)$$

$$\sum_{r \in \Omega} z_r = m \quad (5)$$

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Managing a large number of variables

It is not realistic to be able to give *all* variables together (in a single call) to a general-purpose MILP solver.

The same difficulty appear evens solving the relaxed linear programming problem. We can solve LP with *some* routes. Using the economic interpretation of the dual variable, a route with a negative reduced cost may reduce the cost of the solution.

If α_i is the dual variable of (4) and β is the dual variable of (5), then the **reduced cost** of z_r is $\bar{c}_r = c_r - \beta - \sum_{i \in N} \alpha_i$.

Hence, we need to solve:

Pricing Problem

Find a new variable r minimizing the reduced cost $\bar{c}_r$ associated to given dual variables.

It consists of finding the *Shortest Path Problem with a Resource Constraint* (see e.g. I.Himmich, I.Hallaoui, F.Soumis, A multiphase dynamic programming algorithm for the shortest path problem with resource constraints European Journal of Operational Research (2024) 315, 470–483).

With a **pricing algorithm**, the model can be solved in an iterative *column-generation* framework.

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The same difficulty appear evens solving the relaxed linear programming problem. We can solve LP with *some* routes. Using the economic interpretation of the dual variable, a route with a negative reduced cost may reduce the cost of the solution.

If α_i is the dual variable of (4) and β is the dual variable of (5), then the **reduced cost** of z_r is $\bar{c}_r = c_r - \beta - \sum_{i \in N} \alpha_i$.

Hence, we need to solve:

Pricing Problem

Find a new variable r minimizing the reduced cost $\bar{c}_r$ associated to given dual variables.

It consists of finding the *Shortest Path Problem with a Resource Constraint* (see e.g. I.Himmich, I.Hallaoui, F.Soumis, A multiphase dynamic programming algorithm for the shortest path problem with resource constraints European Journal of Operational Research (2024) 315, 470–483).

With a **pricing algorithm**, the model can be solved in an iterative *column-generation* framework.

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Pricing algorithm for elementary routes

An easy-to-implement alternative to solve the pricing problem is:

Add the following variables to the master problem

$$x_a = \sum_{r \in R} b_a^r z_r \quad \forall a \in A$$

and let $\bar{c}_a$ the reduce cost of x_a . Then

$$\begin{aligned} \min \quad & \sum_{a \in A} \bar{c}_a v_a \\ \sum_{a \in \delta^+(i)} v_a &= \sum_{a \in \delta^-(i)} v_a = w_i \quad \forall i \in N \\ \sum_{a \in \delta^+(0)} v_a &= \sum_{a \in \delta^-(0)} v_a = 1 \\ \sum_{i \in N} q_i w_i &\leq Q \\ u_j &\geq u_i + v_{ij} - (|N| - 1)(1 - v_{ij}) + (|N| - 2)v_{ji} \quad \forall i, j \in N \mid i \neq j \\ v_a &\in \{0, 1\} \quad \forall a \in A \\ w_i &\in \{0, 1\} \quad \forall i \in N. \end{aligned}$$

Branch-and-Cut-and-Price algorithm

$$\min \sum_{a \in A} c_a x_a$$

subject to

$$\sum_{a \in \delta^+(i)} x_a = \sum_{a \in \delta^-(i)} x_a = 1 \quad \forall i \in N$$

$$\sum_{a \in \delta^+(0)} x_a = \sum_{a \in \delta^-(0)} x_a = m$$

$$x_a = \sum_{r \in R} b_a^r z_r \quad \forall a \in A \quad \text{column -- generation}$$

$$\sum_{a \in \delta^+(S)} x_a \geq r(S) \quad \forall S \subset N \quad \text{row -- generation}$$

$$x_a \in \{0, 1\} \quad \forall a \in A.$$

Outline

- 1 Problem description
- 2 Compact formulations
- 3 Cutting-plane formulations
- 4 Set-partitioning formulations
- 5 Comparing formulations**

Computational experiments with E-n22-k4

model	LP obj-val	time	iter	cuts	time	iter	cuts
cut-and-branch	262.0000	0.00			0.02	26	58
VVEHICLE + TMTZ	308.5000	0.01			127.83		
VDECOMP + TMTZ	308.5000	0.02			250.86		
VDECOMP + TPRED	308.5000	0.23			923.94		
VDECOMP + TMCf	318.0000	0.21			286.19		
VPRED	308.5701	0.04			204.68		
VSCF	330.5630	0.01			2.21		
VMTZ	308.5000	0.00			25.69		
VMCF1a	328.0217	0.34			313.52		
VMCF1b	350.0250	0.21			121.79		
VMCF1a-VMCF1b	351.5425	0.32			184.71		
VMCF2a	338.5811	0.61			113.87		
VMCF2b	354.5072	0.36			190.89		
VMCF2a-VMCF2b	356.3716	0.43			156.30		
VMCF2a-VMCF2c	345.4568	0.87			111.26		
VMCF2b-VMCF2c	359.6614	0.78			738.25		
VMCF2a-VMCF2b-VMCF2c	359.6614	0.81			426.01		
VMCF3	359.6614	0.33			92.72		
SEC B&C	318.0000	0.00	4	40			
FCC B&C	353.6000	0.00	40	39	0.27	323	301
RCC B&C	374.5000	0.00	28	77	2.69	152	429
B&P	373.8750	6.68		1116			

opt-obj-val(m -TSP)=318 ; opt-obj-val(CVRP)=375



Further details in

- A.N.Letchford, J.J.Salazar. Projection results for vehicle routing. Mathematical Programming 105 (2006) 251–274.
- A.N.Letchford, J.J.Salazar. Stronger multi-commodity flow formulations of the capacitated vehicle routing problem. European Journal of Operational Research 244 (2015) 730–738.
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Chapter 7 in “Handbook on Transport Modeling”
Edited by Sophie N. Parragh and Tom Van Woensel,
Series Research Handbooks in Transport Studies,
Edward Elgar Publishing Ltd, 2026.

